MG-ACC-70 Access Node Optical Transport Failure SOP (Arista)

Node: MG-ACC-70 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_2695

Description

The Optical Transport Failure on MG-ACC-70 Access Node at MG Road Core is attributed to the degradation of SFP/XFP transceivers, fiber cuts, or Rx/Tx power loss. This failure domain occurs on the Access role due to the node's reliance on optical transport for data transmission. Arista's EOS (Extensible Operating System) utilizes protocols like OSPF and IS-IS for routing, which in turn rely on the optical transport infrastructure for packet forwarding. Hardware-wise, the SFP/XFP transceivers are responsible for converting electrical signals to optical signals, which are then transmitted over the fiber optic cables. In the event of a failure, the SFP/XFP's degradation or the fiber cut can lead to a loss of signal integrity, resulting in ALARM_2695. This failure has a Major severity due to its impact on network availability and throughput.

Pre-requisite Checks

- check: Verify that the MG Road Core site is accessible and secure for remote hands dispatch, status: Completed

- check: Confirm that the redundant link status is up and functional, status: Completed

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cut	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces SFP-1/1/1 detail, Expected Output: SFP-1/1/1 is up, line protocol is up, Description: This command verifies the status of the SFP-1/1/1 interface, which is suspected to be faulty. The expected output indicates that the interface is up and functional, but may not reveal the root cause of the failure.

Command: show interface SFP-1/1/1 statistics, Expected Output: Input errors: 0, output errors: 0, Description: This command provides detailed statistics about the SFP-1/1/1 interface, including input and output errors. A high number of errors may indicate a hardware issue or degradation.

Command: show log | include ALARM_2695, Expected Output: ALARM_2695: Optical transport failure on SFP-1/1/1, Description: This command searches the system log for the ALARM_2695 message, which is triggered
```

```
by the optical transport failure. The expected output indicates that
the failure occurred on the SFP-1/1/1 interface.
```

Resolution Steps

Step 1: Step: Replace the faulty SFP/XFP, Command: replace SFP-1/1/1, Description: This step involves replacing the faulty SFP/XFP with a new one. The replace command will automatically configure the new SFP/XFP and bring the interface up.

Step 2: Step: Verify the redundant link status, Command: show interface SFP-1/1/1 detail, Description: After replacing the SFP/XFP, this step verifies that the redundant link status is up and functional. The show interface command provides detailed information about the interface, including its status and statistics.

Post-Check Verification

```
Command: show interface SFP-1/1/1 detail, Expected Output: SFP-1/1/1
is up, line protocol is up, Description: This command verifies that
the SFP/XFP has been replaced successfully and the interface is up and
functional.

Command: show interface SFP-1/1/1 statistics, Expected Output: Input
errors: 0, output errors: 0, Description: This command verifies that
the SFP/XFP is functioning correctly and there are no errors.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP, Command: replace SFP-1/1/1 original, Description: This step involves restoring the original SFP/XFP. The replace original command will automatically configure the original SFP/XFP and bring the interface up.